

7.3 Software Reuse Processes

NOTE — Users of this International Standard who wish to adopt organizational software reuse practices may supplement these provisions with those of **IEEE Std 1517™-1999**, *IEEE Standard for Information Technology — Software Life Cycle Processes — Reuse Processes*, which provides more detailed activities and tasks for institutionalizing reuse.

7.3.1 Domain Engineering Process

7.3.1.1 Purpose

The purpose of the Domain Engineering Process is to **develop and maintain domain models, domain architectures, and reusable assets for a domain**. This process supports systematic reuse by identifying commonalities and variabilities across a family of systems, subsystems, or applications.

7.3.1.2 Outcomes

As a result of successful implementation of the Domain Engineering Process:

- a) the representation forms for the domain models and domain architectures are selected;
- b) the boundaries of the domain and its relationships to other domains are established;
- c) a domain model that captures the essential common and differing features, capabilities, concepts, and functions in the domain is developed;
- d) a domain architecture describing the family of systems within the domain, including their commonalities and variabilities, is developed;
- e) assets belonging to the domain are specified;
- f) assets belonging to the domain are acquired or developed and maintained throughout their life cycles; and
- g) the domain models and architectures are maintained throughout their life cycles.

NOTE 1 — Domain models and architectures provide the foundation for creating reusable assets for a class of systems, subsystems, or applications.

NOTE 2 — The Domain Engineering Process may overlap with development and maintenance processes that use assets produced by the Domain Engineering Process.

7.3.1.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Domain Engineering Process.

Reference — IEEE Std 1517™ provides a more detailed set of activities and tasks that align with those defined below.

7.3.1.3.1 Process Implementation

This activity consists of the following tasks:

7.3.1.3.1.1

The domain engineer shall create and execute a **Domain Engineering Plan**, describing scope, objectives, methods, responsibilities, and schedules for domain analysis, design, and asset provisioning.

7.3.1.3.1.2

The domain engineer shall select the **representation forms** to be used for domain architectures and models (e.g., feature models, reference architectures, ontologies, UML profiles).

7.3.1.3.1.3

The domain engineer shall establish procedures for **receiving, resolving, and providing feedback** to the asset manager whenever problems or change requests occur for assets developed by the domain engineer.

7.3.1.3.2 Domain Analysis

This activity consists of the following tasks:

7.3.1.3.2.1

The domain engineer shall define the **boundaries of the domain** and the relationships between this domain and other related domains.

7.3.1.3.2.2

The domain engineer shall identify the **current and anticipated needs of stakeholders** of software products within this domain.

7.3.1.3.2.3

The domain engineer shall build the **domain models** using the selected representation forms. These models shall describe essential domain concepts, commonalities, variabilities, and interactions.

7.3.1.3.2.4

The domain engineer shall construct a **vocabulary or ontology** that provides standardized terminology to describe important domain concepts and the relationships among common assets.

7.3.1.3.2.5

The domain engineer shall classify and document the domain models in accordance with the organization's modeling standards and asset acceptance procedures.

7.3.1.3.2.6

The domain engineer shall conduct **domain analysis reviews**, involving software developers, asset managers, domain experts, and users.

7.3.1.3.2.7

The domain engineer shall submit the approved domain models to the **asset manager** for storage, cataloging, and reuse distribution.

7.3.1.3.3 Domain Design

This activity consists of the following tasks:

7.3.1.3.3.1

The domain engineer shall create and document the **domain architecture**, consistent with the domain model and following organizational standards.

7.3.1.3.3.2

For each entity selected for design as a reusable asset, the domain engineer shall develop and document an **asset specification**, including functionality, interfaces, constraints, and variability points.

7.3.1.3.3.3

Each asset specification shall be evaluated in accordance with the organization's **asset acceptance and certification procedures**.

7.3.1.3.3.4

The domain engineer shall conduct **domain design reviews**, involving software developers, domain experts, and asset managers.

7.3.1.3.3.5

The domain engineer shall submit the approved domain architecture to the asset manager for inclusion in the organizational asset base.

7.3.1.3.4 Asset Provision

For each asset developed or acquired, this activity consists of the following tasks:

7.3.1.3.4.1

The domain engineer shall obtain the asset by **acquisition or development**.

7.3.1.3.4.2

The domain engineer shall **document and classify** the asset according to the organization's asset taxonomy (e.g., reusable component, design pattern, domain model, service, library).

7.3.1.3.4.3

The domain engineer shall evaluate the asset in accordance with the organization's **asset acceptance and certification procedures**, including testing, documentation review, and interface conformance.

7.3.1.3.4.4

The domain engineer shall conduct **asset reviews**, involving software developers and asset managers.

7.3.1.3.4.5

The domain engineer shall submit the approved asset to the **asset manager** for cataloging, storage, and controlled distribution.

7.3.1.3.5 Asset Maintenance (Reuse-Related)

The following reuse-related task shall be added to the **Software Maintenance Process (6.4.10)**:

7.3.1.3.5.1

When analyzing requests for asset modification and choosing implementation options, the domain engineer shall consider:

- a) conformance with the domain models and domain architecture;
 - b) impact on the systems and software products that use the asset;
 - c) impact on future users of the asset; and
 - d) impact on the **reusability** of the asset.
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7.3.1.4 Deliverables

The following deliverables shall be produced during this process:

- Domain Engineering Plan;
 - Domain models, vocabularies, and ontologies;
 - Domain architecture documentation;
 - Asset specifications, classification records, and acceptance reports;
 - Domain and asset review records;
 - Asset submission and cataloging records;
 - Maintenance impact assessments for modified assets.
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7.3.1.5 Audit and Configuration Control

All domain engineering artifacts (models, architectures, specifications, and assets) shall be placed under **configuration management**.

Audit evidence, including review records and acceptance reports, shall be retained in accordance with organizational and regulatory requirements.

Changes to domain models or assets shall follow the organization's change control procedures and be fully traceable.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — provides requirements and domain insights for asset modeling.
- **Software Architectural Design (7.1.3)** — consumes domain architectures and reusable design assets.
- **Software Construction (7.1.5)** — reuses domain assets during implementation.
- **Software Configuration Management (7.2.2)** — manages versions and

baselines of reusable assets.

- **Software Quality Assurance (7.2.3)** — ensures assets and domain models meet quality and acceptance standards.
- **Software Verification (7.2.4)** and **Software Validation (7.2.5)** — verify and validate assets and domain architectures prior to reuse.
- **Software Maintenance (6.4.10)** — applies reuse-specific maintenance rules for asset evolution.